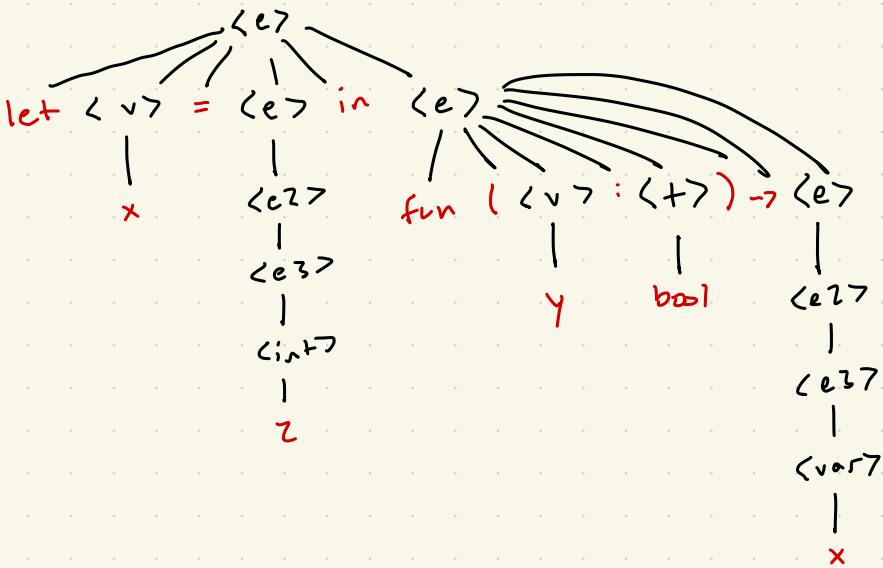
# Mini-Project I Check-in Written Solutions

1.




2.

$$
\dfrac{
\dfrac{}{\emptyset \vdash \text{true} : \text{bool}}\,(\text{true})
\qquad
\dfrac{
\dfrac{\dfrac{}{\{x:\text{bool}\} \vdash 2 : \text{int}}\,(\text{int-lit})
\qquad
\dfrac{\dfrac{}{\{x:\text{int}, y:\text{bool}\} \vdash x : \text{int}}\,(\text{var})}{\{x:\text{int}\} \vdash \text{fun } (y:\text{bool}) \to x : \text{bool} \to \text{int}}\,(\text{fun})}{\{x:\text{bool}\} \vdash \text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x : \text{bool} \to \text{int}}\,(\text{let})
\qquad
\dfrac{}{\{x:\text{bool}\} \vdash x : \text{bool}}\,(\text{var})
}{\{x:\text{bool}\} \vdash (\text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x)\ x : \text{int}}\,(\text{app})
}{\emptyset \vdash \text{let } x = \text{true in } (\text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x)\ x : \text{int}}\,(\text{let})
$$

3. $[2/x](\text{fun } y \to x) = \text{fun } y \to [2/x]x = \text{fun } y \to 2$

$$\dfrac{\dfrac{}{2 \Downarrow 2}\,(\text{intLitE}) \qquad \dfrac{}{\text{fun } y \to 2 \Downarrow \lambda y.\, 2}\,(\text{funE})}{\text{let } x = 2 \text{ in fun } y \to x \Downarrow \lambda y.\, 2}\,(\text{letE})$$

4.

$$\dfrac{\dfrac{}{\emptyset \vdash 2 \Downarrow 2}\,(\text{intLitE}) \qquad \dfrac{}{\{x \mapsto 2\} \vdash \text{fun } (y:\text{bool}) \to x \Downarrow (\{x \mapsto 2\}, \bullet \mapsto \text{fun } (y:\text{bool}) \to x)}\,(\text{funE})}{\emptyset \vdash \text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x \Downarrow (\{x \mapsto 2\}, \bullet \mapsto \text{fun } (y:\text{bool}) \to x)}\,(\text{letE})$$

5.

$$\dfrac{\dfrac{\dfrac{}{\{x \mapsto T\} \vdash 2 \Downarrow 2}\,(\text{intLiteE}) \qquad \dfrac{}{\{x \mapsto 2\} \vdash \text{fun } (y:\text{bool}) \to x \Downarrow (\{x \mapsto 2\}, \cdot \mapsto \text{fun } (y:\text{bool}) \to x)}\,(\text{funE})}{\{x \mapsto T\} \vdash \text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x \Downarrow (\{x \mapsto 2\}, \cdot \mapsto \text{fun } (y:\text{bool}) \to x)}\,(\text{letE}) \qquad \dfrac{}{\{x \mapsto T\} \vdash x \Downarrow T}\,(\text{varE}) \qquad \dfrac{}{\{x \mapsto T, y \mapsto T\} \vdash x \Downarrow T}\,(\text{varE})}{\{x \mapsto T\} \vdash (\text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x)\ x \Downarrow T}\,(\text{appE})$$

$$\dfrac{\dfrac{}{\emptyset \vdash \text{true} \Downarrow T}\,(\text{trueE}) \qquad \{x \mapsto T\} \vdash (\text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x)\ x \Downarrow T}{\emptyset \vdash \text{let } x = \text{true in } (\text{let } x = 2 \text{ in fun } (y:\text{bool}) \to x)\ x \Downarrow T}\,(\text{letE})$$